

Fault-Onset Labeling Bounds Reported Accuracy in the Tennessee Eastman Benchmark: A Leakage-Controlled Study of Six Classical Classifiers

Abstract

Fault diagnosis in the Tennessee Eastman Process (TEP) is a standard benchmark for data-driven process monitoring, yet published results are difficult to compare. We present a leakage-controlled benchmark of six classical classifiers: logistic regression, decision tree, random forest, histogram-based gradient boosting, linear support vector machine and XGBoost. Data were partitioned by complete simulation runs, standardization was fitted on training data only, hyperparameters were selected on a separate validation partition, variability was measured over five seeds, and the frozen model was evaluated once on the complete official test partition of 10.08 million observations. XGBoost ranked first in repeated validation (macro-F1 0.7683 ± 0.0017) and reached a test macro-F1 of 0.7200, accuracy of 0.6754 and MCC of 0.6621 under the labels distributed with the dataset.

These figures are bounded by the benchmark itself. Each fault is injected 160 samples into every 960-sample test run, so literal labeling caps the recall of every fault class at 0.8333; faults 6 and 7 attain 0.8333 and 0.8332, the ceiling itself. Rescoring the same frozen model over the post-onset segment, without refitting, raises macro-F1 to 0.8054, accuracy to 0.7949 and MCC to 0.7852, and the apparent validation-to-test generalization gap of +0.0483 becomes -0.0125 under a symmetric comparison. Across seven classifiers the relative accuracy gain lies between 0.142 and 0.150 against a structural bound of 0.1587, so the effect is a property of the labeling convention rather than of any estimator. Normal operation and faults 3, 9 and 15 remain mutually confounded after the correction, consistent with their known weak observability.

Keywords: Tennessee Eastman Process, fault diagnosis, machine learning, process monitoring, reproducible benchmark, fault-onset labeling

31 1. Introduction

32 Industrial processes operate through interacting physical, chemical, and
33 control mechanisms. A local disturbance can propagate through feedback
34 loops, alter several measured variables, and eventually compromise product
35 quality, equipment integrity, safety, or environmental performance. Fault
36 detection and diagnosis (FDD) therefore constitute central elements of mod-
37 ern process monitoring [1, 2, 3]. Detection determines whether the process
38 has departed from acceptable operation, while diagnosis seeks to identify the
39 underlying condition so that operators or automated systems can respond
40 appropriately.

41 Model-based diagnosis can be effective when reliable first-principles mod-
42 els are available. However, plant-wide systems are frequently nonlinear, high-
43 dimensional, stochastic, and only partially observed. These properties have
44 motivated data-driven monitoring, in which statistical or machine-learning
45 models infer diagnostic boundaries from historical measurements [4, 2]. Clas-
46 sical multivariate methods such as principal-component analysis (PCA), par-
47 tial least squares (PLS), and Fisher discriminant analysis established impor-
48 tant foundations for process monitoring [5]. More recent studies increasingly
49 employ tree ensembles, support vector machines, neural networks, and se-
50 quence models to represent nonlinear relationships and dynamic behavior.

51 The Tennessee Eastman Process (TEP), originally introduced by Downs
52 and Vogel [6], remains one of the most widely used benchmarks in this field.
53 It represents a reactor–separator–recycle chemical process with 41 measured
54 variables, 11 manipulated variables, and a diverse set of disturbances. Its
55 continued value arises from the coexistence of relatively distinctive faults
56 and difficult conditions whose trajectories overlap normal operation. The
57 expanded simulation data published by Rieth et al. [7] provide hundreds
58 of independent runs per condition and non-overlapping random seeds for
59 development and testing, enabling large-scale and reproducible evaluation.

60 Despite extensive use of the TEP, reported results are often difficult to
61 compare. Studies may split temporally adjacent observations rather than
62 complete runs, tune models on the test set, report only accuracy, or omit
63 class-level behavior. Treating autocorrelated observations as independent
64 experimental replicates can produce optimistic estimates, while a high global
65 score can hide failures in specific operating conditions. These issues are es-
66 pecially relevant when the normal state is confused with a fault, because false
67 alarms can reduce operator confidence and generate unnecessary interven-

tions.

This study addresses these limitations through a leakage-controlled benchmark of six classical supervised classifiers: logistic regression, decision tree, random forest, histogram-based gradient boosting, linear support vector machine (SVM), and XGBoost. The study asks five questions:

1. Which classical model provides the strongest macro-averaged diagnostic performance under repeated validation?
2. Does the selected model maintain its performance on a complete, independent official test partition?
3. Which operating conditions account for the remaining diagnostic errors?
4. How much of the reported error is attributable to the fault-onset labeling convention rather than to the classifier?
5. Which model offers the best compromise between diagnostic performance and computational cost?

The main contribution is an auditable protocol that separates model selection from final testing, preserves complete simulation runs, quantifies variability over five seeds, reports confidence intervals, and examines precision, recall, F1, and confusion patterns for all 21 classes. Rather than presenting a single headline score, the analysis identifies a compact ambiguity group that defines the practical limit of the selected classifier.

A second contribution is methodological and, in our view, of wider consequence for the TEP literature. The expanded dataset injects each fault after a fixed warm-up interval, so a fraction of every faulty run is nominally normal yet carries the fault label. This fraction differs between development runs (4%) and official test runs (16.7%). We quantify how much of the reported error and of the apparent generalization gap is produced by this convention alone, and we report every headline metric under both the literal labeling and a post-onset labeling. To our knowledge this decomposition is not routinely reported, which may account for part of the dispersion among published TEP results.

2. Related Work

2.1. Data-driven process monitoring

Data-driven FDD spans latent-variable modeling, classification, clustering, change detection, and temporal modeling. The review by Venkatasubramanian et al. [1] established a broad taxonomy covering quantitative models,

104 qualitative models, and process-history-based methods. Chiang et al. [5, 4]
 105 showed how PCA, discriminant PLS, and Fisher discriminant analysis could
 106 support chemical-process diagnosis. Later reviews emphasized that industrial
 107 data may be nonlinear, dynamic, nonstationary, multimodal, and autocorre-
 108 lated [2, 3]. These properties motivate flexible nonlinear classifiers, but also
 109 demand careful validation.

110 Multivariate statistical monitoring of the TEP has a long history. Raich
 111 and Çinar combined statistical process monitoring with disturbance diagnosis
 112 in multivariable continuous processes [20], and multiscale extensions using
 113 wavelet decompositions were introduced to separate phenomena occurring
 114 at different time scales [22]. A broader survey of plant-wide disturbance
 115 detection is given by Thornhill and Horch [25].

116 A substantial line of work addresses the serial correlation of process data
 117 directly. Dynamic PCA augments the data matrix with time-lagged variables
 118 to capture time-varying structure [35]; canonical variate analysis models the
 119 dynamics explicitly and was compared against PCA and DPCA on the TEP
 120 simulator by Russell et al. [36], later extended with kernel density estimation
 121 for nonlinear dynamics [37]. These methods are relevant here for a reason
 122 beyond their performance: they treat the temporal structure of a run as
 123 the object of analysis, whereas an observation-level classifier treats each row
 124 independently. The labeling question examined in Section 3.2 arises precisely
 125 at that boundary.

126 2.2. Classical machine learning for TEP diagnosis

127 The families evaluated here are standard: classification and regression
 128 trees [8], random forests [9], gradient boosting [10] and its regularized histogram-
 129 based formulation in XGBoost [11], together with multinomial logistic regres-
 130 sion and margin-based linear classification [12] as controls that test whether
 131 approximately linear decision surfaces suffice. All implementations in this
 132 study were based on scikit-learn [13], except XGBoost, which used its refer-
 133 ence implementation [11].

134 Support vector machines have been applied to the TEP in several forms.
 135 Chiang et al. combined Fisher discriminant analysis with support vector
 136 machines for fault diagnosis [23]; Kulkarni et al. incorporated process knowl-
 137 edge into the kernel to detect TEP faults [24]; and kernel extensions of PCA
 138 and independent component analysis were compared against SVM on the
 139 same benchmark [26]. These studies use kernels that the present work could
 140 not afford at the sample size considered here, which is the reason the linear

141 surrogate of Section 3.4 should be read as a lower bound rather than as the
142 best attainable margin classifier.

143 Deep architectures have been applied extensively to the same benchmark.
144 Chadha and Schwung compared feedforward and convolutional networks for
145 TEP fault detection [31]; hybrid designs combining support vector machines
146 with latent-variable preprocessing [34], hierarchical LSTM structures [32],
147 and metaheuristically optimized deep models [33] have all reported strong
148 results. Most directly comparable to the present study, Lomov et al. [30]
149 evaluated recurrent, attention-based and temporal convolutional architec-
150 tures on the same expanded TEP dataset used here. The present work does
151 not propose a competing architecture; it examines a property of the evalua-
152 tion protocol that applies to all of them.

153 *2.3. Evaluation protocols and leakage*

154 Concern about evaluation protocol is not specific to process monitoring.
155 In a survey across disciplines that have adopted machine learning, Kapoor
156 and Narayanan identified leakage in seventeen fields, affecting at least 294
157 publications and in several cases producing conclusions that did not survive
158 correction [29]. Their taxonomy distinguishes eight mechanisms, two of which
159 bear directly on the present benchmark: temporal leakage, where information
160 from the future informs a prediction about the past, and non-independence
161 between training and evaluation samples, where records that share a common
162 origin are distributed across partitions and treated as independent replicates.

163 Both are latent in observation-level TEP classification. Consecutive sam-
164 ples within a simulation run are strongly autocorrelated, so a split performed
165 at the observation level places near-duplicate records on both sides of the par-
166 tition. The run-level protocol of Section 3.3 is designed to avoid this. The
167 labeling question analyzed in this paper is related but distinct: it concerns
168 not how records are divided, but whether the label attached to a record
169 describes the state of the plant at that instant.

170 *2.4. Evaluation beyond accuracy*

171 Accuracy can be misleading when performance varies substantially among
172 classes. Macro-F1 gives equal importance to every class by averaging class-
173 specific F1 scores, while weighted-F1 weights them by support. Balanced
174 accuracy averages class recalls. MCC summarizes agreement between pre-
175 dictions and true labels and remains informative for multiclass problems [14].
176 Confusion matrices are indispensable because they reveal whether errors are

Table 1: Treatment of the pre-onset window in observation-level studies on the Tennessee Eastman benchmark. Three studies on the same simulator adopt three distinct conventions: relabeling the pre-onset segment as normal, retaining it under the fault label, and not addressing it.

Study	Task	Methods	Pre-onset handling	Scored as fault
Yin et al. [27]	binary detection	SVM, PLS	stated, relabeled normal	800
Sun et al. [28]	detection + identif.	Bayesian RNN	stated, retained	960
Agarwal et al. [32]	21-class diagnosis	LSTM-SAE	not mentioned	960
This work	21-class diagnosis	six classical	stated, both protocols	960 / 800

diffuse or concentrated in specific class pairs. The present study therefore uses macro-F1 as the primary model-selection criterion and treats the other metrics as complementary evidence.

2.5. Reported treatment of the fault-onset window

The analysis reported in this paper is only consequential if the treatment of the onset window varies, or goes unstated, across published work on the same benchmark. We therefore examined the data-description and experimental-protocol sections of studies for which the full text was available, recording the diagnostic task, the stated treatment of the pre-onset segment, and the number of test samples scored against the fault label. The purpose is illustrative rather than exhaustive. Three studies on the same simulator adopt three different conventions, which is sufficient to establish that the choice is not standardized; Table 1 reports them.

The question applies to both distributions of the benchmark. In the original data of Downs and Vogel as distributed by Chiang et al. [6, 4], each condition is simulated for 24 hours in training and 48 hours in testing at a three-minute interval, giving 480 and 960 samples, with the fault injected after one hour and eight hours respectively. The pre-onset fractions are therefore 4.2% and 16.7%, essentially identical to those of the expanded dataset analyzed here. The artifact is thus a property of the benchmark design rather than of any particular redistribution of it, and the results of Section 4.5 bound what can be reported on either version.

Yin et al. [27] address the window directly. Their data description states that fault information emerges eight hours after start-up, so that the first 160 test samples appear normal while the remaining 800 are genuinely faulty, and their discussion of the classification output is explicit about the consequence: the target label is -1 over the first 160 samples and $+1$ from the 161st

204 onward. The pre-onset segment is thus relabeled as normal rather than
205 scored against the fault, and their fault-detection rate is computed over 800
206 samples.

207 Sun et al. [28] state the same structure – the simulator runs 160 samples
208 in the normal state before the disturbance is introduced, then continues for
209 800 further samples – but evaluate over all 960. Awareness of the injection
210 instant is therefore not sufficient by itself; what determines comparability is
211 what is done with the segment.

212 Agarwal et al. [32] describe their data as 1440 samples per class, of which
213 the first 480 are used for calibration and the remaining 960 for testing, and
214 identify the source distribution explicitly. The instant of fault injection is
215 not mentioned anywhere in the data description or the experimental proto-
216 col. The consequence is not incidental to that study’s argument: its stated
217 objective is to improve the diagnosis of the incipient faults 3, 9 and 15, to
218 which end pseudo-random binary signals are actively injected into the plant.
219 Under literal labeling, part of the difficulty that motivates such an interven-
220 tion is contributed by observations in which no fault is yet present.

221 The pattern across the three is not one of carelessness, and it admits
222 a structural explanation. In binary detection the pre-onset segment has a
223 natural destination: the normal class already exists in the problem, so rela-
224 beling those observations is the obvious treatment, and the false-alarm rate
225 is in fact defined over precisely that segment. The convention is therefore
226 built into the evaluation. In multiclass diagnosis the same move is no longer
227 neutral, because assigning pre-onset observations of every faulty run to class
228 0 alters the prior of the normal condition and, in the present data, would
229 add 1.6 million observations to a class that already competes with faults 3,
230 9 and 15. Faced with that, the simplest course is to label each run by its
231 fault number throughout, and the distinction disappears without ever being
232 posed as a choice. What is lost in the transition from detection to diagnosis
233 is not information about the data but a convention for handling it.

234 The practical consequence is that reported figures are not commensurable
235 across the three treatments. For a single frozen model the difference between
236 literal and post-onset labeling is 0.0854 in macro-F1 and 0.1231 in MCC (Sec-
237 tion 4.5), larger than the margins that typically separate competing methods
238 on this benchmark. We do not claim a systematic review here; the point is
239 that three conventions coexist, that the choice is seldom identified as such,
240 and that it should be reported alongside the unit of partitioning.

241 **[TBD: Optional: extend Table 1 with further studies as their**

242 full texts are consulted, in particular Yin et al., Lomov et al. and
 243 Chadha & Schwung. Record for each the exact sentence describing
 244 the data, or its absence.]

245 3. Materials and Methods

246 3.1. Tennessee Eastman Process data

247 The TEP simulates a plant-wide reactor–separator–recycle system and
 248 includes 41 process measurements (XMEAS) and 11 manipulated variables
 249 (XMV), totaling 52 predictors [6]. The original problem statement specifies
 250 the plant but not a control structure; the simulations distributed as bench-
 251 mark data use the plant-wide control scheme of Lyman and Georgakis [21],
 252 so the recorded trajectories reflect both the process dynamics and the closed
 253 loops acting on them. The classification task contained 21 labels: class 0 rep-
 254 resented fault-free operation and classes 1–20 represented the predefined fault
 255 conditions. The CSV distribution used in this work is a format conversion
 256 of the expanded TEP simulation data of Rieth et al. [7, 15].

257 The four source files contained 15,330,000 observations and 55 columns:
 258 the 52 process variables plus the identifiers `faultNumber`, `simulationRun`,
 259 and `sample`. Development runs contained 500 observations each. Official
 260 test runs contained 960 observations each. Every class had 500 official test
 261 runs, yielding

$$21 \times 500 \times 960 = 10,080,000 \quad (1)$$

262 test observations. Each class therefore contributed 480,000 observations
 263 to the final test.

264 3.2. Fault-onset windows and labeling protocols

265 The expanded dataset is generated by simulating each condition for 25
 266 hours (development) or 48 hours (official test) with a sampling interval of
 267 three minutes, and by injecting the fault after a fixed warm-up interval: one
 268 hour in faulty development runs and eight hours in faulty test runs [7, 16, 17].
 269 In sample indices, the fault is active from sample 21 in development runs and
 270 from sample 161 in official test runs. Fault-free runs contain no injection.

271 Consequently, the labels distributed with the dataset are only condition-
 272 ally correct. In a faulty development run, samples 1–20 (4.0% of the run)
 273 describe nominal operation while carrying a fault label; in a faulty official

274 test run, samples 1–160 (16.7% of the run) do so. Two consequences follow
 275 directly and are, to our knowledge, seldom made explicit.

276 First, under an observation-level protocol with literal labels, the recall of
 277 every fault class is bounded above by

$$R_c^{\max} = \frac{960 - 160}{960} = 0.8333, \quad c = 1, \dots, 20, \quad (2)$$

278 whenever the classifier assigns pre-onset observations to some class other
 279 than c . No such ceiling applies to class 0, whose test runs contain no injection.
 280 Second, because the mislabeled fraction differs between development (4.0%)
 281 and test (16.7%), a metric computed on the two partitions is not measuring
 282 the same quantity, and part of any observed validation-to-test degradation
 283 is attributable to this asymmetry rather than to generalization.

284 We therefore report all headline metrics under two labeling protocols:

- 285 • **Literal labeling (L)**: every observation retains the label distributed
 286 with the dataset. This is the protocol implicitly adopted by most pub-
 287 lished observation-level TEP results and is retained here for compara-
 288 bility.
- 289 • **Post-onset labeling (P)**: pre-onset observations of faulty runs are
 290 excluded *at evaluation time*, so that each scored observation carries a
 291 label consistent with the physical state of the plant. The model is not
 292 refitted: protocol P rescores the predictions of the same frozen model
 293 that was fitted under protocol L, including its 4% of contaminated
 294 training observations. The comparison L versus P therefore isolates
 295 the labeling convention alone, and the reported post-onset figures are
 296 conservative, since a model additionally trained without the contami-
 297 nated observations could only do better.

298 A third variant, in which pre-onset observations are excluded from fitting
 299 as well, was not evaluated here; it would confound the labeling effect with a
 300 change in the training distribution and is left for future work.

301 Protocol L is used for model selection, so that the frozen-model decision
 302 is unaffected by the present analysis. Protocol P is reported alongside it in
 303 Section 4.5. A third option, relabeling pre-onset observations as class 0, was
 304 not adopted because it alters the class prior of the normal condition and
 305 would confound the analysis of the ambiguity group.

306 3.3. Leakage-controlled partitioning and preprocessing

307 The original development runs were divided at the run level into training
 308 and validation partitions using a fixed random seed and a stratified 80/20
 309 split within each source and class. This produced 8,400 training runs (400
 310 per class), 2,100 validation runs (100 per class), and 10,500 official test runs
 311 (500 per class). No run was assigned to more than one partition.

312 All 52 predictors were standardized using

$$z_{ij} = \frac{x_{ij} - \mu_j^{(\text{train})}}{\sigma_j^{(\text{train})}}, \quad (3)$$

313 where $\mu_j^{(\text{train})}$ and $\sigma_j^{(\text{train})}$ were estimated exclusively from the 4.2 million
 314 training observations. The fitted transformation was then applied unchanged
 315 to validation and test data. This procedure prevented information from the
 316 validation or test partitions from influencing preprocessing.

317 3.4. Candidate classifiers

318 Six model families were compared. Hyperparameter tuning evaluated
 319 three candidates per family, producing 18 configurations. The selected con-
 320 figurations were subsequently frozen:

- 321 • **Logistic regression:** $C = 1.0$, maximum 1,000 iterations;
- 322 • **Decision tree:** unrestricted depth and minimum leaf size 1;
- 323 • **Random forest:** 400 trees, unrestricted depth, 50% of features con-
 324 sidered per split, and minimum leaf size 1;
- 325 • **HistGradientBoosting:** learning rate 0.05, 350 boosting iterations,
 326 and at most 31 leaf nodes;
- 327 • **Linear SVM:** stochastic-gradient hinge-loss classifier with $\alpha = 10^{-5}$,
 328 maximum 2,000 iterations, and tolerance 10^{-3} ;
- 329 • **XGBoost:** 250 trees, maximum depth 6, learning rate 0.1, row sub-
 330 sampling 0.8, column subsampling 0.8, multiclass probabilistic objec-
 331 tive, and histogram tree construction.

332 The tuning stage ranked configurations by macro-F1 and used MCC as a
 333 secondary criterion. The official test partition was not accepted as an input
 334 by either the tuning or repeated-validation programs.

335 The linear SVM was implemented as a stochastic-gradient surrogate rather
 336 than as an exact margin solver, because quadratic-programming solvers were
 337 not tractable at this sample size. Its score should therefore be read as a lower
 338 bound on linear-margin performance, and the evidence for nonlinearity re-
 339 ported in Section 5.1 rests primarily on the multinomial logistic regression,
 340 which was optimized to convergence. Kernel SVMs were not evaluated for
 341 the same computational reason.

342 3.5. Repeated validation

343 After selecting the best configuration for each model family, variability
 344 was measured across seeds 2026–2030. For each seed, 40 complete runs per
 345 class were sampled from training and independently from validation. Thus,
 346 every repetition used 420,000 training and 420,000 validation observations
 347 while retaining the full 500-observation trajectories of all selected runs.

348 For each metric, the mean, sample standard deviation, and 95% confi-
 349 dence interval were calculated across the five repetitions. With $n = 5$, the
 350 interval was

$$\bar{x} \pm t_{0.975,4} \frac{s}{\sqrt{5}}. \quad (4)$$

351 Because only five seeds were used, these intervals quantify observed seed-
 352 to-seed variability but should not be interpreted as definitive evidence of
 353 pairwise statistical superiority. In particular, overlapping intervals for the
 354 two leading models were interpreted conservatively.

355 In addition to the interval estimates, the six model families were compared
 356 with the Friedman rank test over the five seeds treated as blocks, followed
 357 by the Nemenyi post-hoc procedure [19]. Pairwise comparisons used the
 358 Wilcoxon signed-rank test with Holm correction. The critical difference of
 359 the Nemenyi procedure was computed as $CD = q_\alpha \sqrt{k(k+1)/6N}$ with $k =$
 360 6 models and $N = 5$ blocks. Results are reported in Section 4.1. With
 361 five blocks these procedures have low power by construction, and they are
 362 reported to document that the prespecified ranking rule was not substituted
 363 for an inferential check, not to establish pairwise superiority.

364 Because a single frozen evaluation cannot express sampling uncertainty,
 365 the official-test metrics were additionally accompanied by 95% intervals ob-

366 tained by resampling complete runs with replacement (1,000 bootstrap repli-
367 cates over the 10,500 test runs). The run, not the observation, is the resam-
368 pling unit, consistent with the argument that temporally adjacent observa-
369 tions are not independent replicates.

370 *3.6. Frozen final evaluation*

371 XGBoost candidate 1 was selected because it achieved the highest repeated-
372 validation macro-F1. The final seed was fixed at 2026 before test evaluation.
373 The final development set combined 40 complete runs per class from train-
374 ing and 40 from validation, totaling 840,000 observations. The frozen model
375 was then evaluated once on all 10,080,000 official test observations, without
376 subsampling or post-test adjustment.

377 Two earlier executions aborted during structural validation, before any
378 model fitting or metric computation, because of an incorrect run-length as-
379 sertion; only that assertion was changed, and the full incident log is available
380 in the repository.

381 The choice of 840,000 development observations, rather than the full
382 5,250,000 available, was imposed by the computational budget. To bound the
383 effect of that choice, the frozen configuration was refitted on $\{10, 20, 40, 80, 160\}$
384 complete runs per class and scored on the full validation partition, sampling
385 at the run level throughout. Results are reported in Section 4.2.

386 *3.7. Performance measures and reproducibility*

387 The reported measures were accuracy, balanced accuracy, precision, re-
388 call, class-specific F1, macro-F1, weighted-F1, MCC, confusion matrix, train-
389 ing time, inference time, inference time per observation, and serialized model
390 size. Zero divisions in class-specific measures were assigned zero. Data and
391 output integrity were monitored with SHA-256 hashes and manifests. The
392 final execution used Python 3.12.13, scikit-learn 1.9.0, and XGBoost 3.4.0 on
393 Linux.

394 **4. Results**

395 *4.1. Model selection under repeated validation*

396 Table 2 summarizes the five-seed validation results. XGBoost ranked first
397 with macro-F1 of 0.7683 ± 0.0017 and MCC of 0.7426 ± 0.0018 . HistGradi-
398 entBoosting was a close second, with macro-F1 of 0.7643 ± 0.0034 . Random
399 forest achieved 0.7398, followed by decision tree at 0.6714. The two linear

Table 2: Repeated-validation performance over five seeds. Macro-F1 was the prespecified primary selection criterion.

Rank	Model	Macro-F1 (mean $\pm$ SD)	95% CI	MCC (mean $\pm$ SD)
1	XGBoost	0.7683 ± 0.0017	[0.7661, 0.7705]	0.7426 ± 0.0018
2	HistGradientBoosting	0.7643 ± 0.0034	[0.7601, 0.7686]	0.7391 ± 0.0045
3	Random Forest	0.7398 ± 0.0021	[0.7372, 0.7424]	0.7159 ± 0.0021
4	Decision Tree	0.6714 ± 0.0013	[0.6699, 0.6730]	0.6537 ± 0.0015
5	Logistic Regression	0.4685 ± 0.0074	[0.4592, 0.4777]	0.4567 ± 0.0073
6	Linear SVM	0.4438 ± 0.0099	[0.4314, 0.4561]	0.4377 ± 0.0109

400 baselines performed substantially worse: logistic regression achieved 0.4685
 401 and linear SVM achieved 0.4438.

402 The Friedman test over the five seeds rejected the hypothesis of equal
 403 performance among the six families ($\chi^2_5 = 24.54$, $p = 1.71 \times 10^{-4}$), with a
 404 Kendall coefficient of concordance of $W = 0.982$. The ordering is thus almost
 405 perfectly consistent from seed to seed: XGBoost obtained the best macro-F1
 406 in four of the five repetitions, and the mean ranks were 1.2 (XGBoost), 1.8
 407 (HistGradientBoosting), 3.0 (random forest), 4.0 (decision tree), 5.0 (logistic
 408 regression) and 6.0 (linear SVM).

409 Post-hoc resolution is nevertheless limited. With $N = 5$ blocks the Ne-
 410 menyi critical difference is 3.372 mean ranks, so only three of the fifteen
 411 pairwise comparisons are individually significant, all of them between a tree
 412 ensemble and a linear baseline. The separation between XGBoost and Hist-
 413 GradientBoosting is 0.6 mean ranks, far below the critical difference. The
 414 Wilcoxon signed-rank test cannot help here either: with five paired observa-
 415 tions its smallest attainable two-sided p -value is 0.0625, so no comparison can
 416 reach $\alpha = 0.05$ even under a perfect win record, and after Holm correction
 417 none does.

418 These two facts are complementary rather than contradictory. The high
 419 concordance shows that the ranking is not an artifact of seed choice; the
 420 wide critical difference shows that five repetitions cannot certify small dif-
 421 ferences. Both support the conservative reading adopted here: the ordering
 422 is reproducible, the gap between the two leading models is not resolvable at
 423 this sample size, and additional seeds would be required to settle it.

424 Figure 1 shows the ranking and its uncertainty. The confidence intervals
 425 of XGBoost and HistGradientBoosting overlapped. Accordingly, XGBoost
 426 was selected by the prespecified macro-F1 rule, but the result is not presented
 427 as proof of a statistically significant difference between the two models. The

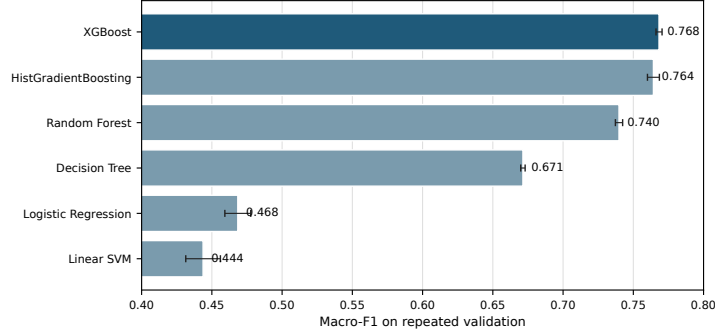


Figure 1: Repeated-validation macro-F1 for the six candidate algorithms. Error bars denote 95% confidence intervals across five seeds.

large separation between nonlinear ensembles and linear baselines indicates that nonlinear interactions are central to this diagnostic task.

4.2. Sensitivity to the size of the development set

Table 3 reports macro-F1 on the complete validation partition as a function of the number of complete runs per class used for fitting, under both labeling protocols. All fits used the frozen configuration and 24 threads.

Table 3: Validation macro-F1 as a function of the training set size, in complete runs per class. Δ is the gain over the preceding row, that is, the effect of doubling the data.

Runs/class	Observations	F1 (L)	Δ	F1 (P)	Δ	Fit (s)
10	105,000	0.7538	—	0.7780	—	59.0
20	210,000	0.7591	0.0052	0.7847	0.0067	65.2
40	420,000	0.7672	0.0082	0.7930	0.0083	111.6
80	840,000	0.7756	0.0083	0.7988	0.0058	175.4
160	1,680,000	0.7806	0.0051	0.8016	0.0028	357.4

Two observations follow. First, the procedure reproduces the repeated-validation result: at 40 runs per class it yields a macro-F1 of 0.7672 against the 0.7683 ± 0.0017 of Table 2, a difference of 0.0011, which confirms that the curve is measured under the protocol of Section 3.5 rather than under a variant of it.

Second, the returns diminish. Doubling the data from 80 to 160 runs per class adds 0.0051 in macro-F1 under literal labeling and 0.0028 under post-onset labeling; quadrupling from the 40 runs per class used in the frozen fit

442 adds 0.0134 and 0.0086 respectively. The frozen model is therefore not at
443 the asymptote, and the figures reported in Section 4.3 are conservative in
444 that sense, but the residual headroom is of the same order as the separation
445 between the two leading model families and does not alter the ranking.

446 The comparison that matters for this paper is between that headroom and
447 the effect under study. Quadrupling the training data from the size actually
448 used gains 0.0086 in post-onset macro-F1; changing the labeling convention
449 on a single fixed model changes macro-F1 by 0.0854, an order of magnitude
450 more. Whatever remains to be gained from more data, it is small beside
451 what is gained or lost by how the pre-onset window is scored.

452 4.3. Official test performance

453 After the protocol was frozen, the selected XGBoost model was fitted to
454 840,000 development observations and evaluated on the complete official test
455 set. Table 4 presents the final metrics under literal labeling. Accuracy and
456 balanced accuracy were both 0.6754, macro-F1 and weighted-F1 were both
457 0.7200, and MCC was 0.6621. The equality of accuracy and balanced accu-
458 racy, and of macro-F1 and weighted-F1, follows from the exactly balanced
459 class supports, and holds only under literal labeling; under the post-onset
460 protocol of Section 4.5 the fault classes retain 400,000 observations while the
461 fault-free class retains 480,000, so the two pairs of metrics separate. Macro-
462 F1 exceeds accuracy because accuracy equals macro-recall under balanced
463 supports, while precision exceeds recall in most classes; the harmonic mean
464 of a high precision and a moderate recall is larger than the recall alone. The
465 gap between macro-F1 and accuracy is therefore itself a signature of the
466 systematic under-recall documented in Section 4.5.

Table 4: Frozen XGBoost performance on the official test set.

Metric	Value
Accuracy	0.6754
Balanced accuracy	0.6754
Macro-F1	0.7200
Weighted-F1	0.7200
MCC	0.6621
Training time (s)	170.34
Inference time (s)	56.57
Inference (ms/observation)	0.005612
Development observations	840,000
Test observations	10,080,000

Relative to repeated validation, macro-F1 decreased by 0.0483 and MCC by 0.0805. Three mechanisms could account for this difference: independent random states in the test simulations, the longer duration of test runs, and the larger fraction of mislabeled pre-onset observations. The third mechanism is both the largest and the only one that is purely a labeling artifact. Under literal labeling the attainable recall per fault class falls from $480/500 = 0.96$ in development runs to $800/960 = 0.8333$ in test runs, a ratio of 0.868, which is of the same order as the observed degradation. Section 4.5 shows that the gap does not merely shrink under post-onset labeling but changes sign: the post-onset test macro-F1 of 0.8054 exceeds the literal-label validation figure of 0.7683 by 0.0371.

That comparison alone would not be symmetric, because the validation figure was itself computed under literal labels and carries its own 4% pre-onset contamination. The validation partition was therefore rescored under both protocols as well. Applying the selected XGBoost configuration to all 1,050,000 validation observations gave a literal-label macro-F1 of 0.7706, which agrees with the five-seed repeated-validation figure of 0.7683 to within 0.0023 and confirms that this partition is a faithful reference. Excluding the 40,000 pre-onset observations, 3.81% of the partition, raised macro-F1 to 0.7953, accuracy from 0.7556 to 0.7843, and MCC from 0.7440 to 0.7739.

The correction on the validation partition is thus +0.0246 in macro-F1, against +0.0854 on the test partition, a ratio of 0.29 that tracks the ratio of contaminated fractions (3.81% against 15.87%). Transferring the measured validation correction to the repeated-validation reference gives a post-

onset validation macro-F1 of approximately 0.7929, against a post-onset test macro-F1 of 0.8054. Table 5 summarizes the decomposition.

Table 5: Validation-to-test difference in macro-F1 under both labeling protocols. The gap present under literal labeling does not survive a symmetric comparison.

Protocol	Validation	Test	Difference
Literal (L)	0.7683	0.7200	+0.0483
Post-onset (P)	0.7929	0.8054	−0.0125

Under a symmetric comparison the gap is not merely reduced but reversed, and its magnitude falls to roughly a quarter of the literal-label value, within the range of the seed-to-seed variability of Table 2. We therefore conclude that the reported generalization gap is an artifact of the labeling convention rather than evidence of degradation on unseen trajectories, and that the literal-label figure of 0.7200 understates the discriminative capacity of the frozen model by 0.0854 in macro-F1.

Two caveats attach to this conclusion. The validation rescoring used the tuned configuration fitted to the full training partition, not the five per-seed refits of the repeated-validation procedure, so the transferred correction is an estimate of the effect rather than an exact recomputation; the close agreement between 0.7706 and 0.7683 bounds the error of that substitution. Furthermore, the pre-onset destinations differ between partitions: 61.1% of pre-onset validation observations were assigned to the group $\{0, 3, 9, 15\}$ against 92.0% in the test partition, which is consistent with the development segment being only 20 samples long and overlapping the simulation transient.

4.4. Class-specific performance

Figure 2 and Table 6 demonstrate marked heterogeneity. Seventeen classes achieved F1 scores from 0.7429 to 0.9080. Classes 6 and 7 were strongest, with F1 of 0.9080 and 0.9075, respectively. Classes 1, 2, 5, and 14 also approached or exceeded 0.8958.

Four classes were persistently difficult: class 9 (F1=0.1635), class 15 (0.1683), normal operation (class 0; 0.1718), and class 3 (0.2357). Class 3 had the highest recall within this group (0.4218) but very low precision (0.1636), indicating systematic overprediction.

A pattern in the recall column deserves emphasis. Classes 6 and 7 reached recalls of 0.8333 and 0.8332, which coincide with the structural ceiling of

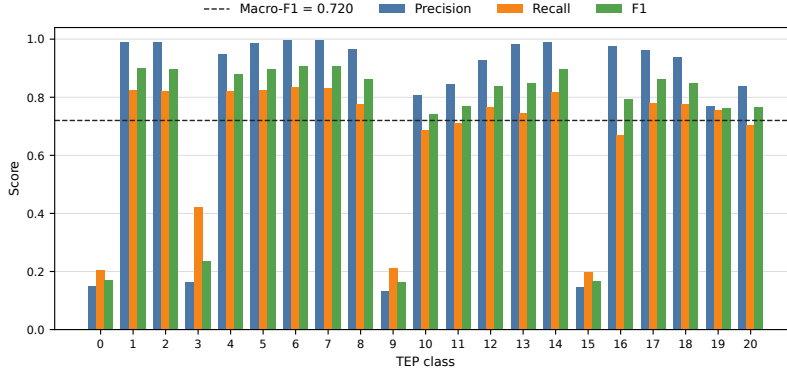


Figure 2: Class-wise precision, recall, and F1 on the complete official test partition. The dashed line indicates overall macro-F1. Class 0 is normal operation.

Eq. (2) to within one part in ten thousand. Classes 1, 2, 4, 5 and 14 lie between 0.8187 and 0.8243, that is, at 98–99% of that ceiling. The frozen model is therefore essentially saturated on these classes once the pre-onset segment is removed, and their reported F1 values of roughly 0.88–0.91 are limited by the labeling convention rather than by the classifier.

4.5. Effect of the fault-onset window

Table 7 compares the frozen model under the two labeling protocols of Section 3.2. No refitting, retuning or reselection was performed; the same frozen model and the same predictions are scored under two label conventions.

Excluding the 1,600,000 pre-onset observations, 15.87% of the partition, raised macro-F1 by 0.0854, accuracy by 0.1195 and MCC by 0.1231. No model was refitted and no hyperparameter was changed.

Two diagnostics establish that this is a labeling effect rather than a favourable choice of subset. First, the frozen model assigned 16.10% of the pre-onset observations to class 0 and 75.86% to classes 3, 9 or 15, that is, 92.0% to the group of conditions that are indistinguishable from nominal operation, and only 4.20% to the fault label they nominally carry. The model therefore identifies the pre-onset segment as a period in which nothing detectable is occurring, which is correct, and literal labeling scores this as error. Second, restricting the evaluation to post-onset observations raised the recall of the 17 well-separated classes to between 0.8015 and 0.9999, the upper value corresponding to essentially complete recovery of faults 6 and 7, while the

Table 6: Class-wise results for the frozen XGBoost model on the official test set. Every class had support of 480,000 observations.

Class	Precision	Recall	F1	Class	Precision	Recall	F1
0	0.1481	0.2045	0.1718	11	0.8447	0.7095	0.7712
1	0.9894	0.8243	0.8993	12	0.9271	0.7669	0.8395
2	0.9896	0.8209	0.8974	13	0.9828	0.7444	0.8471
3	0.1636	0.4218	0.2357	14	0.9889	0.8187	0.8958
4	0.9477	0.8222	0.8805	15	0.1465	0.1978	0.1683
5	0.9879	0.8234	0.8982	16	0.9756	0.6681	0.7931
6	0.9973	0.8333	0.9080	17	0.9624	0.7809	0.8622
7	0.9964	0.8332	0.9075	18	0.9367	0.7760	0.8488
8	0.9665	0.7779	0.8620	19	0.7701	0.7559	0.7630
9	0.1328	0.2126	0.1635	20	0.8391	0.7029	0.7650
10	0.8066	0.6885	0.7429				

Table 7: Frozen XGBoost under literal (L) and post-onset (P) labeling on the official test partition. Intervals are 95% run-level bootstrap intervals over 1,000 replicates.

Metric	Literal (L)	Post-onset (P)
Accuracy	0.6754 [0.6716, 0.6797]	0.7949 [0.7898, 0.8006]
Balanced accuracy	0.6754 [0.6748, 0.6760]	0.8006 [0.7998, 0.8013]
Macro-F1	0.7200 [0.7192, 0.7208]	0.8054 [0.8044, 0.8063]
MCC	0.6621 [0.6579, 0.6666]	0.7852 [0.7798, 0.7910]
Observations	10,080,000	8,480,000

ambiguity group improved only marginally (F1 of 0.2217, 0.3238, 0.2100 and 0.2106 for classes 0, 3, 9 and 15, against 0.1718, 0.2357, 0.1635 and 0.1683 under literal labels). The ambiguity group is therefore not a labeling artifact; the ceiling on the remaining seventeen classes is.

The two phenomena are nevertheless connected. Class 3 was predicted 1,237,562 times against a true support of 480,000, an excess of 757,562 observations. A substantial part of that excess consists of pre-onset segments belonging to other faults, which the model routes to the weakly observable conditions because, at those instants, nothing has yet happened. The over-prediction of class 3 is thus partly a consequence of the labeling convention and partly a consequence of genuine overlap, and only the post-onset analysis separates the two.

Table 8: Effect of post-onset labeling across model families. The last column reports the relative accuracy gain $\Delta A/A_P$, whose structural upper bound is $1 - 8,480,000/10,080,000 = 0.1587$.

Model	Macro-F1 (L)	Macro-F1 (P)	$\Delta A/A_P$
Decision tree	0.6330	0.7167	0.148
Extra trees	0.6965	0.7786	0.148
Random forest	0.7081	0.7912	0.149
HistGradientBoosting	0.7128	0.7972	0.149
XGBoost [†]	0.7200	0.8054	0.150
Logistic regression	0.4283	0.4698	0.142
Linear SVM	0.4207	0.4655	0.142

[†]Frozen model of Table 4, complete test partition.

555 This distinction matters for comparability. As documented in Section 2.5,
556 published observation-level results on the expanded TEP dataset do not con-
557 sistently state whether pre-onset samples were retained, discarded, or rela-
558 beled, yet the three choices are separated by several points of macro-F1 for an
559 identical model. Two studies reporting different scores may therefore differ
560 in labeling convention rather than in method. We recommend that the treat-
561 ment of the onset window be reported explicitly as part of the experimental
562 protocol, in the same way that the unit of partitioning is reported.

563 4.6. Generality of the labeling effect across model families

564 If the effect of Section 4.5 is a property of the labeling convention rather
565 than of the selected classifier, it should appear in every model evaluated on
566 the same partition. To test this, an auxiliary experiment was rescored un-
567 der both protocols. That experiment predates the frozen benchmark, uses
568 a per-class subsample of the test partition, and includes an extremely ran-
569 domized trees model in place of XGBoost; its absolute levels are therefore
570 not comparable with Table 4, and only the differences between protocols are
571 interpreted. No model was refitted.

572 The five tree-based models gained between 0.0821 and 0.0854 in macro-
573 F1, a spread of 0.003 across estimators whose capacities differ by orders of
574 magnitude, from a single unpruned tree to a regularized boosting ensemble.
575 No overfitting mechanism produces that uniformity, because the degradation
576 associated with overfitting scales with model capacity.

577 The relative accuracy gain explains the pattern quantitatively. If a clas-
 578 sifier assigns pre-onset observations to any class other than the labeled fault,
 579 its literal accuracy is its post-onset accuracy scaled by the retained fraction
 580 of the partition, so that $\Delta A/A_P$ is bounded above by 0.1587 independently
 581 of the model. The seven models fall between 0.142 and 0.150, the residual
 582 being the small fraction of pre-onset observations that match the literal la-
 583 bel by chance. The linear baselines gain less in absolute terms, 0.0415 and
 584 0.0448, simply because they start lower; in relative terms they are affected
 585 almost identically. They also route far fewer pre-onset observations to the
 586 group $\{0, 3, 9, 15\}$, 20.6% and 41.9% against 65.7–70.2% for the ensembles,
 587 and are penalized in the same proportion regardless.

588 The consequence is that the effect is not a property of any estimator.
 589 Any observation-level result reported on this dataset under literal labels un-
 590 derstates the post-onset discriminative capacity of the model by a factor
 591 predictable from its own accuracy. Comparisons between published results
 592 are correspondingly affected whenever the treatment of the onset window
 593 differs or is unstated.

594 4.7. *Structure of the remaining errors*

595 The normalized confusion matrix in Fig. 3 shows that errors were not
 596 uniformly distributed. They concentrated in the four-class group $\{0, 3, 9, 15\}$.
 597 Among true class-9 observations, 32.39% were predicted as class 3, 20.24%
 598 as class 0, and 18.08% as class 15. Normal observations were assigned to
 599 classes 3, 9, and 15 in 32.02%, 21.51%, and 18.24% of cases, respectively. For
 600 class 15, the corresponding proportions were 30.96%, 20.53%, and 20.03%
 601 for predicted classes 3, 9, and 0.

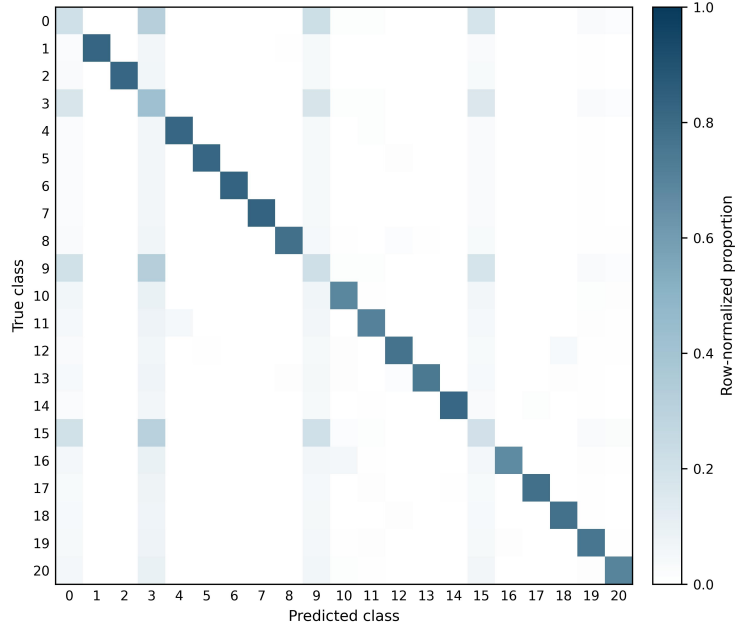


Figure 3: Row-normalized confusion matrix for the frozen XGBoost model on the official test partition.

Class 3 was predicted 1,237,562 times despite having 480,000 true observations. This excess explains its low precision and demonstrates that the model used class 3 as a frequent destination for ambiguous trajectories. The persistence of the same difficult group during repeated validation and official testing suggests an intrinsic overlap under an observation-level representation rather than an isolated sampling artifact.

4.8. Computational performance

Table 9 reports the cost of all six families under identical conditions during repeated validation, which is required to answer the performance–cost question and cannot be inferred from the selected model alone. Training times are reported as means over the five seeds and are not directly comparable with the 170.34s of Table 4, which was measured during the final evaluation on twice as many observations. The degree of parallelism was an invocation-time argument of each stage rather than a fixed setting, and it was not recorded in the run manifests, so the absolute times of the two stages reflect different thread counts on a 24-core host. The comparison between

Table 9: Performance and computational cost of the six families under repeated validation (mean over five seeds, 420,000 training observations per repetition). Times measured on the environment of Section 3.7.

Model	Macro-F1	MCC	Train (s)	Inference (μ s/obs)	Size (MB)
XGBoost	0.7683	0.7426	864.7	7.03	18.9
HistGradientBoosting	0.7643	0.7391	73.5	29.40	16.9
Random Forest	0.7398	0.7159	477.3	20.29	8703.8
Decision Tree	0.6714	0.6537	54.0	0.24	28.8
Logistic Regression	0.4685	0.4567	65.4	0.15	0.005
Linear SVM	0.4438	0.4377	4.1	0.16	0.006

families within Table 9 is unaffected, because those six runs shared a single invocation, but the cross-stage comparison is not meaningful and no conclusion is drawn from it. Recording the degree of parallelism alongside the timing is a reproducibility requirement that this benchmark did not initially meet, and it has been added to the released code.

Final XGBoost training required 170.34 seconds. Prediction over 10.08 million observations required 56.57 seconds, corresponding to 0.005612 ms per observation or approximately 178,000 observations per second. The serialized final model occupied approximately 6.8 MB. These values indicate that the selected classifier is computationally feasible for high-throughput offline analysis and potentially for online scoring, although end-to-end deployment latency must also include data acquisition, preprocessing, communication, and decision logic.

5. Discussion

5.1. Nonlinearity and model ranking

The ranking provides strong empirical evidence that the TEP classification boundary is not adequately represented by the evaluated linear models. XGBoost, HistGradientBoosting, and random forest substantially outperformed logistic regression and linear SVM even though all models received the same standardized predictors and run-level partitions. Tree ensembles can represent threshold effects and high-order interactions without requiring an explicit feature expansion, which is consistent with the nonlinear and feedback-driven nature of the TEP.

The difference between XGBoost and HistGradientBoosting was small: 0.0040 in mean macro-F1, with overlapping 95% confidence intervals. Thus,

the defensible conclusion is that XGBoost won under the prespecified ranking rule, not that it was conclusively superior under all resampling regimes. Table 9 makes the practical consequence explicit, and it does not favour a single model. HistGradientBoosting trained 11.8 times faster than XGBoost (73.5s against 864.7s) for a deficit of 0.0040 in macro-F1 with overlapping intervals, so it dominates whenever models are retrained frequently or the training budget is constrained. The ordering reverses at inference time, where XGBoost required $7.03\ \mu\text{s}$ per observation against $29.40\ \mu\text{s}$, a factor of 4.2 in favour of XGBoost. Neither model dominates the other: the choice follows from whether the deployment is retraining-bound or scoring-bound, and both conclusions are invisible if only the selected model is profiled.

Model size adds a third and largely independent constraint. The random forest required approximately 8.7 GB, some 460 times the XGBoost artifact, for 0.0285 less macro-F1. Whatever its statistical merits, it is not a candidate for embedded or edge deployment, and this would not be apparent from predictive metrics alone. The two linear baselines occupied roughly 5 kB and inferred in $0.15\ \mu\text{s}$ per observation, which quantifies precisely what is being paid for the nonlinear gain of about 0.30 in macro-F1.

5.2. *Why global performance is insufficient*

The official macro-F1 of 0.7200 summarizes useful performance, but it does not describe a uniformly reliable 21-class diagnostic system. If only macro-F1 or accuracy had been reported, the strong results for 17 classes would obscure the near-failure of normal operation and faults 3, 9, and 15. In operational terms, low precision for normal operation and extensive exchange between normal and faulty labels can increase false alarms and missed interventions.

This result illustrates why class-level precision, recall, F1, and confusion structure should accompany aggregate measures in industrial diagnosis. It also supports a distinction between *benchmark discrimination* and *deployment readiness*. The present model is a strong benchmark classifier for most faults, but additional safeguards are necessary before it could support autonomous decisions.

5.3. *Implications for method selection in industrial practice*

The results above matter for an engineer specifying a diagnostic system, and they matter in a way that is easy to miss. Benchmark figures circulate as evidence of capability: a vendor or an internal team reporting a macro-F1

679 of 0.80 on the TEP appears to offer a better method than one reporting 0.72.
680 Sections 4.5 and 4.6 show that this particular difference is fully accounted
681 for by the labeling convention, for an identical model. The question to put
682 to a reported figure is therefore not only which algorithm produced it, but
683 over which observations it was computed: whether the interval between the
684 disturbance entering the process and its becoming observable was scored as
685 faulty, as normal, or excluded.

686 The benchmark’s fixed injection instant has no counterpart in an op-
687 erating plant, but the underlying interval does. A disturbance entering a
688 unit does not appear simultaneously in the instrumentation: it propagates
689 through transport delay, is attenuated or temporarily masked by the control
690 loops acting to reject it, and becomes visible only when it exceeds the nor-
691 mal variability of the affected measurements. **[CO-AUTHORS: Give a
692 concrete order of magnitude from operating experience, and name
693 what governs it in practice – sensor placement, loop tuning, trans-
694 port lag, sampling and scan rate.]** During that interval a diagnostic
695 model receives data that carries no signature of the condition it is expected
696 to report. Whatever the label attached to those observations in a historical
697 dataset, the model cannot separate them, and a figure of merit computed over
698 them measures the labeling convention rather than the diagnostic system.

699 The group $\{0, 3, 9, 15\}$ makes the same point in a sharper form. These
700 conditions survive the correction of Section 4.5 because they leave no persis-
701 tent signature in the 52 measured variables, and the literature has reported
702 them as weakly detectable for two decades. For a deployment this is not
703 a modeling deficiency to be solved with a better classifier: it is a statement
704 about what the existing instrumentation can observe. **[CO-AUTHORS: From
705 practice: when a condition proves undiagnosable from the available
706 process data, what is the usual course – additional or relocated
707 instrumentation, a different measurement principle, or accepting
708 that the condition is handled by other means? What governs that
709 decision economically?]**

710 Finally, the cost results of Table 9 show that no model dominates. Hist-
711 GradientBoosting trained 11.8 times faster than XGBoost for a difference of
712 0.0040 in macro-F1, while XGBoost inferred 4.2 times faster; the random for-
713 est required approximately 8.7 GB. Which of these constraints binds depends
714 on the deployment rather than on the benchmark. **[CO-AUTHORS: From
715 experience: in the installations you work with, what is the actual
716 limiting factor – how often models are retrained, the latency budget**

717 within the scan cycle, or the memory and storage available on the
718 controller, edge device or historian? An example of a constraint
719 that ruled out an otherwise adequate method would be valuable
720 here.]

721 The common thread is that benchmark performance and deployment
722 readiness are distinct properties, and that the distance between them is not
723 only a matter of model quality. It also depends on how the evaluation was
724 constructed, on what the instrumentation can observe, and on the resources
725 available where the model will actually run.

726 5.4. Interpretation of the ambiguity group

727 The identity of the difficult group is not a new finding, and it would be
728 misleading to present it as one. Faults 3, 9 and 15 have been reported as
729 weakly detectable since the earliest systematic studies of the TEP: moni-
730 toring statistics based on PCA, discriminant PLS and Fisher discriminant
731 analysis report missed-detection rates close to those of nominal operation
732 for exactly these conditions [5, 4], and later comparative studies reproduce
733 the pattern across a wide range of data-driven statistics [18]. The contribu-
734 tion here is the scale and the protocol: the same group is recovered under
735 run-level partitioning, on 10.08 million independent test observations, and it
736 persists after the labeling artifact of Section 4.5 is removed. This strengthens
737 the interpretation that the overlap is a property of the measured variables
738 rather than of any particular estimator.

739 The mechanism, however, constrains which remedies can work. If these
740 disturbances leave no persistent signature in the 52 measured variables, then
741 temporal aggregation over a run reduces the variance of an estimate that is
742 already centered on an ambiguous region: it will sharpen the decision without
743 moving it toward the correct class. Run-level aggregation should therefore
744 be expected to help the 17 separable classes, whose pre-onset transients are
745 transient, far more than the group $\{0, 3, 9, 15\}$, and this asymmetry is itself a
746 testable prediction. A hierarchical design that isolates the ambiguous group
747 and applies a discriminator with explicit memory to classes 0, 3, 9 and 15
748 is the more promising direction, but its plausibility rests on whether any
749 signature exists at all; establishing this is a question of observability, not
750 of classifier capacity, and it is better addressed with residual-based or first-
751 principles analysis than with additional supervised tuning.

752 Both proposals arise from test-set diagnosis and must therefore be eval-
753 uated under a newly defined development/test protocol; the present official

754 test must not be reused for tuning them.

755 5.5. *Threats to validity*

756 Several limitations delimit the conclusions. First, TEP is a high-fidelity
757 simulation benchmark, not a physical plant. Results may not transfer directly
758 to sensor drift, maintenance changes, unmodeled disturbances, communica-
759 tion losses, or evolving operating modes in real facilities. Second, the bench-
760 mark evaluated pointwise multiclass diagnosis and did not measure detection
761 delay, time-to-alarm, false alarms per run, or sustained alarm logic. Third,
762 five seeds provide a useful but small basis for uncertainty estimation. The
763 resulting t intervals characterize the implemented experiment rather than all
764 possible data partitions.

765 Fourth, hyperparameter search was intentionally limited to three can-
766 didates per model family to preserve computational feasibility and trans-
767 parency. A larger search might improve individual algorithms, particularly
768 the linear baselines or random forest. Fifth, the final model was fitted to
769 840,000 of the 5,250,000 available development observations, and model se-
770 lection was carried out at 420,000. Section 4.2 shows that the residual head-
771 room at that size is 0.0086 in post-onset macro-F1 under quadrupling, so
772 the reported figures are conservative; a ranking established at one training
773 scale need not be invariant to scale, and the curve bounds but does not elimi-
774 nate this concern. Sixth, the linear SVM was a stochastic-gradient surrogate
775 and its result should not be read as the best attainable linear-margin per-
776 formance. Seventh, the degree of parallelism was not recorded in the run
777 manifests, so the training times of Table 9 support ordinal comparison be-
778 tween families, which shared one invocation, but not comparison against the
779 timings of other stages. Eighth, interpretation focused on output behavior
780 and did not include feature attribution or causal process analysis. Finally,
781 the dataset had balanced class support, whereas real industrial faults are rare
782 and highly imbalanced. Metrics, thresholds, and alarm costs would require
783 recalibration under field prevalence.

784 6. Conclusion

785 This work presented a reproducible classical-machine-learning benchmark
786 for 21-class fault diagnosis in the Tennessee Eastman Process. Run-level

787 splitting, train-only standardization, separate tuning and validation, five-
788 seed repetition, and a single frozen official test evaluation were used to reduce
789 leakage and post-selection bias.

790 XGBoost achieved the highest repeated-validation macro-F1 ($0.7683 \pm$
791 0.0017) and produced an official test macro-F1 of 0.7200, accuracy of 0.6754,
792 and MCC of 0.6621 over 10.08 million observations under literal labeling,
793 rising to a macro-F1 of 0.8054, an accuracy of 0.7949 and an MCC of 0.7852
794 once the pre-onset segment of each faulty run is excluded, with no refitting of
795 any kind. The validation partition, rescored under the same protocol, moved
796 from 0.7706 to 0.7953, so that the apparent generalization gap of +0.0483 in
797 macro-F1 becomes -0.0125 once the comparison is made symmetric. Seven-
798 teen classes were diagnosed with F1 between 0.7429 and 0.9080, and several
799 of them were shown to sit at the structural recall ceiling imposed by the la-
800 beling convention rather than at the limit of the classifier. Normal operation
801 and faults 3, 9, and 15 remained strongly confounded, in agreement with
802 two decades of TEP monitoring literature, and this confounding survives the
803 removal of the labeling artifact.

804 Three conclusions follow. Nonlinear tree ensembles provide effective di-
805 agnosis for most TEP conditions. Aggregate scores overstate practical reli-
806 ability when a small group of classes dominates the remaining errors. And
807 a benchmark result on the expanded TEP dataset is not interpretable, nor
808 comparable across studies, unless the treatment of the fault-onset window
809 is stated explicitly. A single frozen model differs by 0.085 in macro-F1 and
810 0.123 in MCC between two defensible labeling conventions, which is larger
811 than most of the improvements reported between competing architectures on
812 this benchmark. The effect is moreover common to every model family tested,
813 with a relative accuracy gain between 0.142 and 0.150 across seven classifiers,
814 so it displaces published results without reordering them only when all stud-
815 ies happen to adopt the same convention. We therefore recommend that the
816 treatment of the onset window be stated as part of the experimental protocol
817 in any observation-level study using these data.

818 Future work should investigate temporal aggregation, hierarchical diag-
819 nosis, probability calibration, detection delay, and cost-sensitive alarm rules
820 under a new untouched test protocol. External validation with physical in-
821 dustrial data will be necessary before deployment-oriented conclusions can
822 be drawn.

823 Data and Code Availability

824 The expanded Tennessee Eastman simulation data are publicly available
825 from Harvard Dataverse [7]. The CSV conversion used in this study is avail-
826 able through Kaggle [15]. All code, configuration files, run-level split mani-
827 fests, frozen hyperparameters, five-seed validation outputs, final predictions,
828 class-level metrics, confusion matrices, SHA-256 data hashes, and the en-
829 vironment specification are deposited at an anonymized repository for peer
830 review, **[TBD: anonymous DOI]**; the non-anonymized repository is cited
831 in the accepted version. The analysis of the fault-onset window can be re-
832 produced from the deposited predictions without refitting any model.

833 CRediT Authorship Contribution Statement

834 **First author:** Conceptualization, Investigation, Validation, Writing –
835 original draft, Writing – review & editing. **Second author:** Investigation,
836 Validation, Writing – original draft, Writing – review & editing. **Third**
837 **author:** Conceptualization, Methodology, Software, Formal analysis, Data
838 curation, Writing – original draft, Visualization, Supervision, Project admin-
839 istration.

840 Declaration of Competing Interest

841 The authors declare that they have no known competing financial inter-
842 ests or personal relationships that could have appeared to influence the work
843 reported in this paper.

844 References

- 845 [1] V. Venkatasubramanian, R. Rengaswamy, K. Yin, and S. N. Kavuri,
846 “A review of process fault detection and diagnosis: Part I: Quantitative
847 model-based methods,” *Computers & Chemical Engineering*, vol. 27, no.
848 3, pp. 293–311, 2003. doi: 10.1016/S0098-1354(02)00160-6.
- 849 [2] S. Yin, S. X. Ding, X. Xie, and H. Luo, “A review on basic data-
850 driven approaches for industrial process monitoring,” *IEEE Transac-*
851 *tions on Industrial Electronics*, vol. 61, no. 11, pp. 6418–6428, 2014. doi:
852 10.1109/TIE.2014.2301773.

- 853 [3] C. Ji and W. Sun, “A review on data-driven process monitoring methods:
854 Characterization and mining of industrial data,” *Processes*, vol. 10, no.
855 2, art. 335, 2022. doi: 10.3390/pr10020335.
- 856 [4] L. H. Chiang, E. L. Russell, and R. D. Braatz, *Fault Detection and*
857 *Diagnosis in Industrial Systems*. London, UK: Springer, 2001. doi:
858 10.1007/978-1-4471-0347-9.
- 859 [5] L. H. Chiang, E. L. Russell, and R. D. Braatz, “Fault diagnosis in chem-
860 ical processes using Fisher discriminant analysis, discriminant partial
861 least squares, and principal component analysis,” *Chemometrics and*
862 *Intelligent Laboratory Systems*, vol. 50, no. 2, pp. 243–252, 2000. doi:
863 10.1016/S0169-7439(99)00061-1.
- 864 [6] J. J. Downs and E. F. Vogel, “A plant-wide industrial process control
865 problem,” *Computers & Chemical Engineering*, vol. 17, no. 3, pp. 245–
866 255, 1993. doi: 10.1016/0098-1354(93)80018-I.
- 867 [7] C. A. Rieth, B. D. Amsel, R. Tran, and M. B. Cook, “Additional Ten-
868 nessee Eastman Process simulation data for anomaly detection evalua-
869 tion,” Harvard Dataverse, V1, 2017. doi: 10.7910/DVN/6C3JR1.
- 870 [8] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, *Classification*
871 *and Regression Trees*. Belmont, CA, USA: Wadsworth, 1984.
- 872 [9] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, pp. 5–32,
873 2001. doi: 10.1023/A:1010933404324.
- 874 [10] J. H. Friedman, “Greedy function approximation: A gradient boosting
875 machine,” *The Annals of Statistics*, vol. 29, no. 5, pp. 1189–1232, 2001.
876 doi: 10.1214/aos/1013203451.
- 877 [11] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting sys-
878 tem,” in *Proceedings of the 22nd ACM SIGKDD International Confer-*
879 *ence on Knowledge Discovery and Data Mining*, 2016, pp. 785–794. doi:
880 10.1145/2939672.2939785.
- 881 [12] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*,
882 vol. 20, pp. 273–297, 1995. doi: 10.1007/BF00994018.

- 883 [13] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal*
884 *of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- 885 [14] B. W. Matthews, “Comparison of the predicted and observed secondary
886 structure of T4 phage lysozyme,” *Biochimica et Biophysica Acta*, vol.
887 405, no. 2, pp. 442–451, 1975. doi: 10.1016/0005-2795(75)90109-9.
- 888 [15] A. Melo, “Tennessee Eastman Process CSV: Process simulation
889 data for anomaly detection evaluation,” Kaggle Dataset, accessed
890 Aug. 2026. [Online]. Available: <https://www.kaggle.com/datasets/afnrniomelo/tep-csv>.
891
- 892 [16] C. A. Rieth, B. D. Amsel, R. Tran, and M. B. Cook, “Issues and advances
893 in anomaly detection evaluation for joint human-automated systems,” in
894 *Advances in Human Factors in Robots and Unmanned Systems* (AHFE
895 2017), Cham: Springer, 2018, pp. 52–63.
- 896 [17] C. Reinartz, M. Kulahci, and O. Ravn, “An extended Tennessee Eastman
897 simulation dataset for fault-detection and decision support systems,”
898 *Computers & Chemical Engineering*, vol. 149, art. 107281, 2021. doi:
899 10.1016/j.compchemeng.2021.107281.
- 900 [18] S. Yin, S. X. Ding, A. Haghani, H. Hao, and P. Zhang, “A compar-
901 ison study of basic data-driven fault diagnosis and process monitor-
902 ing methods on the benchmark Tennessee Eastman process,” *Jour-
903 nal of Process Control*, vol. 22, no. 9, pp. 1567–1581, 2012. doi:
904 10.1016/j.jprocont.2012.06.009.
- 905 [19] J. Demšar, “Statistical comparisons of classifiers over multiple data sets,”
906 *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.
- 907 [20] A. Raich and A. Çinar, “Statistical process monitoring and disturbance
908 diagnosis in multivariable continuous processes,” *AIChE Journal*, vol.
909 42, no. 4, pp. 995–1009, 1996.
- 910 [21] P. R. Lyman and C. Georgakis, “Plant-wide control of the Tennessee
911 Eastman problem,” *Computers & Chemical Engineering*, vol. 19, no. 3,
912 pp. 321–331, 1995.

- 913 [22] M. Misra, H. H. Yue, S. J. Qin, and C. Ling, “Multivariate process mon-
914 itoring and fault diagnosis by multi-scale PCA,” *Computers & Chemical*
915 *Engineering*, vol. 26, no. 9, pp. 1281–1293, 2002.
- 916 [23] L. H. Chiang, M. E. Kotanchek, and A. K. Kordon, “Fault diagnosis
917 based on Fisher discriminant analysis and support vector machines,”
918 *Computers & Chemical Engineering*, vol. 28, no. 8, pp. 1389–1401, 2003.
- 919 [24] A. Kulkarni, V. K. Jayaraman, and B. D. Kulkarni, “Knowledge incor-
920 porated support vector machines to detect faults in Tennessee Eastman
921 process,” *Computers & Chemical Engineering*, vol. 29, no. 10, pp. 2128–
922 2133, 2005.
- 923 [25] N. F. Thornhill and A. Horch, “Advances and new directions in plant-
924 wide disturbance detection and diagnosis,” *Control Engineering Prac-*
925 *tice*, vol. 15, no. 10, pp. 1196–1206, 2007.
- 926 [26] Y. W. Zhang, “Enhanced statistical analysis of nonlinear process using
927 KPCA, KICA and SVM,” *Chemical Engineering Science*, vol. 64, no. 5,
928 pp. 801–811, 2009.
- 929 [27] S. Yin, X. Gao, H. R. Karimi, and X. Zhu, “Study on support vec-
930 tor machine-based fault detection in Tennessee Eastman process,” *Ab-*
931 *stract and Applied Analysis*, vol. 2014, art. 836895, 8 pages, 2014. doi:
932 10.1155/2014/836895.
- 933 [28] W. Sun, A. R. C. Paiva, P. Xu, A. Sundaram, and R. D. Braatz, “Fault
934 detection and identification using Bayesian recurrent neural networks,”
935 *Computers & Chemical Engineering*, vol. 141, art. 106991, 2020. doi:
936 10.1016/j.compchemeng.2020.106991.
- 937 [29] S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis
938 in machine-learning-based science,” *Patterns*, vol. 4, no. 9, art. 100804,
939 2023. doi: 10.1016/j.patter.2023.100804.
- 940 [30] I. Lomov, M. Lyubimov, I. Makarov, and L. E. Zhukov, “Fault detection
941 in Tennessee Eastman process with temporal deep learning models,”
942 *Journal of Industrial Information Integration*, vol. 23, art. 100216, 2021.
943 doi: 10.1016/j.jii.2021.100216.

- 944 [31] G. S. Chadha and A. Schwung, "Comparison of deep neural network ar-
 945 chitectures for fault detection in Tennessee Eastman process," in *Proc.*
 946 *22nd IEEE International Conference on Emerging Technologies and*
 947 *Factory Automation (ETFA)*, 2017, pp. 1–8.
- 948 [32] P. Agarwal et al., "Hierarchical deep LSTM for fault detection and diag-
 949 nosis for a chemical process," *Processes*, vol. 10, no. 12, art. 2557, 2022.
 950 doi: 10.3390/pr10122557.
- 951 [33] C. Anitha et al., "Fault diagnosis of Tennessee Eastman process with
 952 detection quality using IMVOA with hybrid DL technique in IIoT," *SN*
 953 *Computer Science*, vol. 4, no. 5, art. 458, 2023. doi: 10.1007/s42979-
 954 023-01851-9.
- 955 [34] X. Gao and J. Hou, "An improved SVM integrated GS-PCA fault di-
 956 agnosis approach of Tennessee Eastman process," *Neurocomputing*, vol.
 957 174, pp. 906–911, 2016.
- 958 [35] W. Ku, R. H. Storer, and C. Georgakis, "Disturbance detection and
 959 isolation by dynamic principal component analysis," *Chemometrics and*
 960 *Intelligent Laboratory Systems*, vol. 30, no. 1, pp. 179–196, 1995.
- 961 [36] E. L. Russell, L. H. Chiang, and R. D. Braatz, "Fault detection in in-
 962 dustrial processes using canonical variate analysis and dynamic principal
 963 component analysis," *Chemometrics and Intelligent Laboratory Systems*,
 964 vol. 51, no. 1, pp. 81–93, 2000.
- 965 [37] P. P. Odiowei and Y. Cao, "Nonlinear dynamic process monitoring using
 966 canonical variate analysis and kernel density estimations," *IEEE Trans-*
 967 *actions on Industrial Informatics*, vol. 6, no. 1, pp. 36–45, 2010.